Improved zero-dimensional qudit stabilizer codes of odd length from circulant graphs

Claude Opus 5.5 Extra Effort[1]

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We report four qudit stabilizer codes, $[\![19,0,9]\!]_5$, $[\![25,0,11]\!]_5$, $[\![29,0,12]\!]_5$, and $[\![21,0,10]\!]_7$, each of which raises the best known lower bound on the minimum distance in Grassl's tables of quantum codes by one. All four are circulant graph codes, i.e., stabilizer states of weighted circulant graphs over $\mathbb{F}_q$. We argue that odd-length zero-dimensional entries are comparatively weak for a structural reason: $\mathbb{F}_{q^2}$-linear Hermitian self-dual codes, which underlie the systematic quasi-twisted searches that populate the qudit tables, exist only at even length, whereas additive graph codes exist at every length. The minimum distances were certified by an exact Brouwer–Zimmermann-type enumeration over disjoint qudit information sets, with every certificate repeated under a different choice of information sets. For $[\![19,0,9]\!]_5$ an independent exhaustive enumeration confirms $d = 9$ and finds exactly $A_9 = 12\,920$ stabilizer elements of minimum weight. The codes were found by an autonomous multi-agent language-model search that operated against a frozen, test-validated verifier, and we describe the safeguards used to exclude evaluator errors. The tabulated upper bounds coincide with the quantum Singleton bound, so all four entries remain open; reaching them would give absolutely maximally entangled stabilizer states.

I. INTRODUCTION

Quantum error-correcting codes of the stabilizer type [1, 2] extend naturally from qubits to qudits of prime-power dimension [3–5]. A stabilizer code with parameters $[\![n,k,d]\!]_q$ encodes k qudits of dimension q into n and detects all errors of weight less than d. For given n, k, and q, the largest achievable d is unknown in general. Grassl's tables [6] record the best known lower and upper bounds for qubits up to $n = 256$ and, since 2024, for $q \in \{3,4,5,7,8\}$ up to $n = 100$. The qudit tables rest mainly on quasi-twisted codes from nearly self-orthogonal $\mathbb{F}_{q^2}$-linear codes [7] and on quantum twisted codes [8]. Any explicit code whose minimum distance exceeds a tabulated lower bound is a new record, and the claim can be checked exactly.

This paper concerns zero-dimensional codes ($k = 0$). An $[\![n,0,d]\!]_q$ code is a stabilizer state whose nonidentity stabilizer elements all have weight at least d. Such states are equivalent to self-dual additive codes over $\mathbb{F}_{q^2}$ and, up to local Clifford operations, to graph states of weighted graphs [9, 11–14]. They are of interest both as resource states and as building blocks for codes with $k > 0$.

Inspection of the qudit tables shows a parity effect at $k = 0$. For example, the table lists $[\![18,0,10]\!]_5$ (meeting the quantum Singleton bound) but only $8 \le d \le 10$ for $n = 19$. Similarly, $[\![24,0,11]\!]_5$ sits next to $10 \le d \le 13$ at $n = 25$, and $[\![28,0,12]\!]_5$ next to $11 \le d \le 15$ at $n = 29$. For $q = 7$, $[\![20,0,10]\!]_7$ sits next to $9 \le d \le 11$ at $n = 21$. Since the generic propagation rule $[\![n,k,d]\!]_q \to [\![n+1,k,d]\!]_q$ requires $k > 0$ [2], a good code at even length does not automatically yield one at the next odd length.

We close part of this gap. Table I lists four circulant graph codes at odd lengths, each improving the previous lower bound by one. Section II recalls the graph-code formalism and explains the parity effect. Section III presents the codes. Section IV describes the exact verification and its validation. Section V describes how the codes were found, and Section VI discusses open problems.

TABLE I. New codes. "Previous" gives the lower and upper bounds in Ref. [6] (snapshot of 10 October 2026, re-checked online 11 October 2026). The upper bounds equal the quantum Singleton bound $\lfloor n/2 \rfloor + 1$. c is the first half of the circulant's first row (Sec. III); "deg" is the vertex degree of the graph.

Code	Previous	New d	deg	$c_1 c_2 \cdots c_m$
$[\![19,0,9]\!]_5$	8–10	9	16	`413014221`
$[\![25,0,11]\!]_5$	10–13	11	20	`323341240420`
$[\![29,0,12]\!]_5$	11–15	12	28	`31214214324143`
$[\![21,0,10]\!]_7$	9–11	10	18	`4416064533`

II. PRELIMINARIES

A. Qudit stabilizer codes

Let q be prime. A Pauli operator on n qudits is specified up to phase by $(\boldsymbol{x}|\boldsymbol{z}) \in \mathbb{F}_q^{2n}$, corresponding to $\bigotimes_i X^{x_i} Z^{z_i}$. Two operators commute if and only if the symplectic form $\langle (\boldsymbol{x}|\boldsymbol{z}), (\boldsymbol{x}'|\boldsymbol{z}') \rangle_s = \boldsymbol{x}\cdot\boldsymbol{z}' - \boldsymbol{z}\cdot\boldsymbol{x}'$ vanishes modulo q. A stabilizer code is defined by an isotropic subspace $S \subseteq \mathbb{F}_q^{2n}$ of dimension $n-k$. Its normalizer is the symplectic complement $N = S^{\perp_s}$, and its minimum distance is $d = \min\{\mathrm{wt}(v) : v \in N \setminus S\}$ for $k > 0$ and $d = \min\{\mathrm{wt}(v) : v \in S \setminus \{0\}\}$ for $k = 0$. Here $\mathrm{wt}(\boldsymbol{x}|\boldsymbol{z}) = |\{i : (x_i, z_i) \neq (0,0)\}|$ is the symplectic weight. The quantum Singleton bound $n - k \ge 2(d-1)$ [4, 17] gives $d \le \lfloor n/2 \rfloor + 1$ for $k = 0$. Equality holds precisely when every reduced state on $\lfloor n/2 \rfloor$ qudits is maximally mixed, i.e., when the stabilizer state is absolutely maximally entangled (AME) [18–20].

B. Graph codes

Let A be a symmetric $n \times n$ matrix over $\mathbb{F}_q$ with zero diagonal, the weighted adjacency matrix of a graph. The rows of $(I \,|\, A)$ span an isotropic subspace of dimension n, because $\langle (e_u|A_u), (e_v|A_v)\rangle_s = A_{vu} - A_{uv} = 0$. The corresponding stabilizer state (graph state) has generators $K_v = X_v \prod_u Z_u^{A_{vu}}$, and its stabilizer elements are $(\boldsymbol{u} \,|\, \boldsymbol{u}A)$ for $\boldsymbol{u} \in \mathbb{F}_q^n$. Hence

$$d = \min_{\boldsymbol{u} \neq 0} \big|\mathrm{supp}(\boldsymbol{u}) \cup \mathrm{supp}(\boldsymbol{u}A)\big|. \tag{1}$$

Every stabilizer state is local-Clifford equivalent to a graph state [10, 11, 13], so graph codes realize every attainable $[\![n, 0, d]\!]_q$.

A graph is *circulant* if $A_{ij} = a_{(j-i) \bmod n}$. Symmetry requires $a_j = a_{n-j}$, so for odd $n = 2m+1$ the first row is $(0, c_1, \ldots, c_m, c_m, \ldots, c_1)$ and there are q^m circulant graph codes of length n. Danielsen [14] searched them exhaustively for alphabets $\mathbb{F}_{m^2}$ with $m \leq 5$. For $\mathbb{F}_{25}$ ($q = 5$) the search extends to $n = 18$, where it attains $d = 10$; the length-19 row of that table has no entry for $\mathbb{F}_{25}$. Circulant constructions are also a standard source of good qubit codes [15, 16].

C. Why odd lengths are weak

A $\mathbb{F}_{q^2}$-linear code that is Hermitian self-dual has dimension $n/2$ and therefore requires even n. Such codes give $[\![n, 0, d]\!]_q$ stabilizer codes, and families of $\mathbb{F}_{q^2}$-linear codes (constacyclic, quasi-cyclic, quasi-twisted) are the backbone of systematic searches [7]. At odd length the corresponding code must be genuinely additive ($\mathbb{F}_q$-linear but not $\mathbb{F}_{q^2}$-linear). Graph codes are additive and exist at every length. This asymmetry offers a plausible explanation for the parity effect noted in the Introduction, and it suggested where an additive family could improve the tables.

III. RESULTS

Each code in Table I is the circulant graph code over $\mathbb{F}_q$ whose adjacency matrix has first row

$$(0,\, c_1, \ldots, c_m,\, c_m, \ldots, c_1), \qquad n = 2m+1, \tag{2}$$

with row i the cyclic right shift of row 0 by i places. The stabilizer is generated by $K_v = X_v \prod_u Z_u^{A_{vu}}$, $v = 0, \ldots, n-1$. Full first rows and explicit stabilizer elements of minimum weight are given in Appendix A. The minimum-weight elements show that the distances are attained, i.e., the certified lower bounds are tight for these particular codes.

The four codes are invariant under the cyclic shift $v \mapsto v+1$. Since the generator K_v has weight $1 + \deg v$, a graph code with distance d needs every vertex degree to be at least $d-1$. Danielsen showed more: $d-1$ equals the minimum vertex degree over the local-complementation orbit of the graph [14]. Our graphs are dense, with degrees 16–28. For $[\![19, 0, 9]\!]_5$ we enumerated all stabilizer elements of weight at most 9 (Sec. IV D). There are none of weight below 9 and exactly

$$A_9 = 12\,920 = 76 \times 170 \tag{3}$$

of weight 9. The factorization is a consistency check. Since $n = 19$ is prime, every nonzero element lies in an orbit of size 19×4 under cyclic shifts and nonzero scalars.

Propagation. A pure $[\![n, 0, d]\!]_q$ code yields $[\![n-1, 1, d-1]\!]_q$ by puncturing [2, 4]. For the four codes this gives $[\![18, 1, 8]\!]_5$, $[\![24, 1, 10]\!]_5$, $[\![28, 1, 11]\!]_5$, and $[\![20, 1, 9]\!]_7$, none of which exceeds the existing table entries (9, 10, 11, and 9, respectively). The improvements are therefore confined to the four $k = 0$ entries.

Status of the entries. The upper bounds in Table I equal the quantum Singleton bound. Every entry thus remains open. The entry $[\![21, 0, d]\!]_7$ now has $d \in \{10, 11\}$, and a code with $d = 11$ would be an AME$(21, 7)$ stabilizer state. Analogous statements hold for $q = 5$ at $n = 19$, 25, and 29.

Search heuristics. Let $E = \sum_{w=1}^{d-1} \binom{n}{w} (q^2-1)^w / q^n$ be the expected number of nonzero stabilizer elements of weight below d in a uniformly random stabilizer state. In a circulant code these elements occur in orbits of size $n(q-1)$ when n is prime, and of size at most $n(q-1)$ in general. Treating the number of such orbits as Poisson with mean $\mu = E/[n(q-1)]$ suggests a success probability of roughly $e^{-\mu}$ per random circulant (Table II). For $[\![19, 0, 9]\!]_5$ the estimate, $e^{-5.9} \approx 2.7 \times 10^{-3}$, matches the observed rate: one in 386 screened samples. This first-moment argument is only a guide, but it predicted correctly which odd-length entries were within reach of random circulant sampling.

TABLE II. First-moment heuristic for random symmetric circulant graph codes at the target distance. q^m is the number of such codes.

Code	q^m	E	$\mu = E/[n(q-1)]$	$e^{-\mu}$
$[\![19, 0, 9]\!]_5$	2.0×10^6	449	5.9	2.7×10^{-3}
$[\![25, 0, 11]\!]_5$	2.4×10^8	714	7.1	7.9×10^{-4}
$[\![29, 0, 12]\!]_5$	6.1×10^9	290	2.5	8.2×10^{-2}
$[\![21, 0, 10]\!]_7$	2.8×10^8	722	5.7	3.2×10^{-3}

IV. VERIFICATION

Because the result is a claim about a minimum over $q^n - 1$ stabilizer elements, we treated verification as the primary deliverable. All checks below are exact, and floating-point arithmetic is never used.

A. Validity and dimension

The verifier reads the generators, checks that all entries lie in $\mathbb{F}_q$, and confirms isotropy pairwise. It computes $k = n - \operatorname{rank} S$ by Gaussian elimination over $\mathbb{F}_q$ and the normalizer as the kernel of $(-Z \mid X)$, checking $\dim N = n + k$.

B. Exact minimum distance

Classical Brouwer–Zimmermann enumeration [21] uses several disjoint information sets, i.e., coordinate sets on which a generator matrix can be brought to systematic form. For symplectic weight the relevant coordinates are qudits rather than individual columns, as in the algorithm of White and Grassl for additive codes [22]. We use the following variant. For information set j we reduce a generator matrix of N (here $N = S$) to systematic form with all pivot columns inside a set I_j of qudits, which is disjoint from the previous sets. Rows are grouped by the qudit carrying their pivot; a qudit carries one or two pivots. The f_j rows without a pivot in I_j form singleton groups. Every pivot column vanishes in all other rows. Hence, if the coefficient vector of a codeword v is nonzero on a groups, then v is nonzero on at least $a - f_j$ qudits of I_j.

Suppose all codewords whose coefficient vectors activate at most t_j groups have been enumerated for each j. Because the I_j are disjoint, every codeword not yet seen satisfies

$$\operatorname{wt}(v) \;\geq\; \sum_j \max\bigl(0,\; t_j + 1 - f_j\bigr). \tag{4}$$

The enumeration raises the t_j in rounds and stops when the bound (4) reaches the weight of the lightest codeword found outside S (outside $\{0\}$ for $k = 0$). That weight is then exact. Two refinements reduce the work. The first active group's local value is normalized to fix scalar multiples. Qudits on which S contains a weight-one element are removed beforehand, which leaves d unchanged for $k > 0$. Each witness codeword reported by the engine is re-checked independently: it must commute with all generators, lie outside S, and have the stated weight.

Table III lists the certification runs. For $k = 0$ codes of length n the algorithm uses two disjoint information sets of about $n/2$ qudits each, so the lower bound grows by about two per round. Each certificate was repeated with a different random choice of information sets, and the results agreed.

C. Validation of the verifier

Before any search began, the verifier was frozen and tested. It reproduces the distances of standard codes: the $[\![5, 1, 3]\!]_2$, Steane, and Shor codes (the last correctly reported as impure), a Golay-derived $[\![23, 1, 7]\!]_2$ code, and five-qudit codes over $\mathbb{F}_3$ and $\mathbb{F}_5$. It rejects invalid inputs, such as noncommuting generators, wrong k, entries outside $\mathbb{F}_q$, and nonprime q. On 500 random stabilizer codes over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ with $n \leq 10$, some padded with fixed qudits, it agrees with a brute-force checker that shares no code with it. That checker enumerates Pauli operators by increasing weight and tests commutation and stabilizer membership directly. On 14 best-known codes stored with Ref. [6] ($q \in \{2, 3, 5, 7\}$, up to $[\![60, 2, 16]\!]_2$), it recovers exact distances equal to the tabulated lower bounds. The sources and the compiled binary were then pinned by cryptographic hash. The submission pipeline refused to accept improvements from a modified verifier. A verdict exceeding a tabulated (proven) upper bound would have been flagged as a verifier fault rather than a discovery.

TABLE III. Exact certification. "Words" is the number of codewords enumerated up to scalars; times are for 6–10 threads on one laptop CPU (Apple M4). Each code was certified twice with different information sets.

Code	Words	Time (s), run 1 / run 2	Result
$[\![19, 0, 9]\!]_5$	2.4×10^7	0.05 / 0.06	$d = 9$
$[\![25, 0, 11]\!]_5$	3.6×10^9	11.5 / 7.3	$d = 11$
$[\![29, 0, 12]\!]_5$	1.7×10^{11}	559 / 384	$d = 12$
$[\![21, 0, 10]\!]_7$	1.2×10^{10}	28.5 / 25.4	$d = 10$

D. Independent check of $[\![19, 0, 9]\!]_5$

For graph codes, Eq. (1) yields a proof that needs no information sets. Every stabilizer element $(\boldsymbol{u}|\boldsymbol{u}A)$ has weight at least $\operatorname{wt}(\boldsymbol{u})$. Hence any element of weight at most w has $\operatorname{wt}(\boldsymbol{u}) \leq w$. A separate program, written independently of the main engine, enumerated all 1 475 748 943 coefficient vectors $\boldsymbol{u}$ with $1 \leq \operatorname{wt}(\boldsymbol{u}) \leq 8$ (first nonzero entry fixed to 1). It found minimum weight 9, which proves $d \geq 9$. Extending the enumeration to $\operatorname{wt}(\boldsymbol{u}) \leq 9$ produced the count $A_9 = 12\,920$. For the three larger codes this direct enumeration would cover between about 10^{11} and 10^{14} coefficient vectors and was not performed. Their distances rest on the two engine certificates.

E. Novelty

On 11 October 2026 the online tables [6] still listed lower bounds of 8, 10, 11, and 9 for the four entries. The previous records for $[\![19, 0, 8]\!]_5$, $[\![25, 0, 10]\!]_5$, and $[\![21, 0, 9]\!]_7$ are recorded as stored generator matrices. The record for $[\![29, 0, 11]\!]_5$ is derived by shortening a $[\![30, 0, 12]\!]_5$ code. Danielsen's exhaustive circulant search [14] stops at $n = 18$ for $\mathbb{F}_{25}$ and does not treat $\mathbb{F}_{49}$. Our literature search found no later construction with these parameters, but

we cannot rule out codes in sources we did not examine.

V. HOW THE CODES WERE FOUND

The codes were found by an autonomous multi-agent system built on a large language model (Claude Sonnet 5.5, Anthropic). It was set up to test whether open, verifiable problems of this type can be attacked by AI agents without compromising rigor. Five agents ran in parallel as non-interactive coding sessions on one laptop. They coordinated through a shared message board, where they could claim targets, post progress and dead ends, and submit candidates. The agents were assigned different families: qubit quasi-cyclic and cyclic codes, qubit graph codes and seeded local search, qutrit codes, and $q = 5, 7$ codes. The pool of targets comprised 4 295 table entries whose improvement by one could be certified in about ten minutes.

The design principle was *verifier first*. Agents could not modify the verifier, the bound tables, or the board software. A submission was copied read-only and verified, and the verdict was posted by the verifier, not by the submitting agent. Only verifier-posted improvements counted.

The agent assigned to $q = 5, 7$ noted the parity effect of Sec. II C. It formulated the first-moment estimate of Table II, selected odd-length $k = 0$ entries with small μ, and wrote a sampler for random symmetric circulants. The sampler used a cheap low-weight prefilter followed by a randomized upper-bound screen. The first certified improvement, $[\![19, 0, 9]\!]_5$, was posted six minutes after launch, and the remaining three followed within 13 minutes. Each came from a single sampling run of a few minutes. The other agents' searches, which included exhaustive qubit circulant graph codes for $26 \le n \le 45$, all Hermitian dual-containing cyclic $\mathbb{F}_4$-linear codes of odd length $41 \le n \le 99$, and exhaustive $\mathbb{F}_3$ circulant, bordered-circulant, and negacirculant graph codes for $14 \le n \le 20$ (circulant up to 23), matched but did not beat the tabulated bounds within certifiable range. Every result in this paper was then re-certified and independently checked as described in Sec. IV.

VI. OUTLOOK

The parity effect suggests that further odd-length $k = 0$ entries in the $q = 5$ and $q = 7$ tables can be improved by additive constructions. By the heuristic of Table II, $[\![31, 0, 13]\!]_5$ ($\mu \approx 9$) is a candidate. Its exact certification is estimated at about three quarters of an hour with the present algorithm. For larger lengths certification rather than search becomes the bottleneck. Exploiting the cyclic automorphism group in the enumeration could reduce the cost by roughly a factor of n for circulant codes. Closing the four entries requires either codes meeting the Singleton bound, which would be AME stabilizer states, or improved upper bounds from linear or semidefinite programming. Even-length entries such as $[\![20, 0, d]\!]_5$ ($9 \le d \le 11$), which lies below the $n = 18$ entry, also merit attention. Finally, the cyclic $\mathbb{F}_4$ search produced qubit candidates at $n = 85$, 91, and 93 (for example of type $[\![91, 1, d]\!]_2$ with tabulated $d \ge 21$) for which the randomized screens found no logical operator below the tabulated bound. A screen is not a proof, and certifying these candidates is far beyond the present enumeration, so we report them only as unverified leads.

ACKNOWLEDGMENTS

We thank M. Grassl for maintaining the code tables on which this work relies. The codes were discovered by an autonomous multi-agent system built on Claude Sonnet 5.5 (Anthropic). The verification software, the analysis, and this manuscript were produced by Claude Opus 5.5 (Anthropic).

DATA AVAILABILITY

The generators of all four codes are given in Table I and Appendix A. Candidate files, the verifier (including its test suite), the independent checker, and the search scripts are available from the authors.

Appendix A: Code data and minimum-weight elements

Full first rows $(a_0, \ldots, a_{n-1})$ of the circulant adjacency matrices, and one stabilizer element of minimum weight $(\boldsymbol{x}|\boldsymbol{z})$ with $\boldsymbol{z} = \boldsymbol{x}A$ for each code, written as strings of $\mathbb{F}_q$ digits:

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TABLE IV. First rows of the adjacency matrices and minimum-weight stabilizer elements.

Code	
$[\![19, 0, 9]\!]_5$	$a =$ 0413014221122410314
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 0200010000001310010 | 0000010000000404122
$[\![25, 0, 11]\!]_5$	$a =$ 0323341240420024042143323
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 3000240110242000000040003 | 4000232100320000000030001
$[\![29, 0, 12]\!]_5$	$a =$ 0312142143241433414234124121 3
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 0402301010003100000000420004 | 0304102000014400000000114000
$[\![21, 0, 10]\!]_7$	$a =$ 044160645333546061 44
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 600060041600500000020 | 000060012002500500310

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